

Text Revision by On-the-Fly Representation Optimization

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Abstract

Text revision refers to a family of natural language generation tasks, where the source and target sequences share moderate resemblance in surface form but differentiate in attributes, such as text style transfer (Shen et al., 2017), text simplification (Xu et al., 2016), counterfactual debiasing (Zmigrod et al., 2019), grammar error correction (Sun et al., 2022) and sentence fusion (Malmi et al., 2019).

As the most popular solution, sequence-to-sequence (seq2seq) learning achieves state-of-the-art results on many text revision tasks today. However, it becomes less applicable when there is no large-scale annotated parallel data for training.

With recent breakthroughs in self-supervised learning have enabled the pre-trained Transformer models (Vaswani et al., 2017), such as BERT (Devlin et al., 2018), RoBERTa (Liu et al., 2019) and GPT (Radford et al., 2020), to learn sufficient distributed representation of natural language, which is universally transferable to a wide range of downstream tasks even without labeled data (Tenney et al., 2019; Zhang et al., 2019; Wu et al., 2020). In this work, we borrow the power of a pre-trained Transformer for text revision without any parallel data.

In this paper, we propose OREO, a method of On-the-fly Representation Optimization for text revision. Instead of generating an entire sequence of tokens from scratch, OREO first detects partial text span to be edited, then conducts in-place span revision:

Step 1: Representation optimization Given an input sentence $X^{(i)}$ at the i -th iteration, RoBERTa parameterized by θ transforms it to a sequence of hidden states $H^{(i)}$, conditioned on which the attribute head estimates the probability of target attribute $P_{W_{\text{Att}}}(z^*|H^{(i)})$. Then, for each revision, we find a small local perturbation on $H^{(i)}$ that maximally increases the likelihood of target attribute. As such, the update rule of

hidden states is:

$$H^{(i+1)} = H^{(i)} - \lambda \frac{\nabla_{H^{(i)}} \mathcal{L}}{\|\nabla_{H^{(i)}} \mathcal{L}\|_2}, \quad (1)$$

where λ is a hyper-parameter that controls the norm of perturbation, and

$$\mathcal{L} = -\log P_{W_{\text{Att}}}(z^*|H^{(i)}). \quad (2)$$

Step 2: Span replacement After hidden states are updated, OREO conducts span replacement. We calculate magnitude of $\nabla_{H^{(i)}} \mathcal{L}$ for i -th token, where $\mathcal{L}$ is calculated with (2), and select the span with largest magnitude. The selected span $X_{t:t+N}^{(i)}$ of length N is replaced by [LM-MASK] tokens. RoBERTa takes as input the masked sequence, and predicts a new span autoregressively with the previously updated hidden states.

The training for OREO is simple: we fine-tune the RoBERTa model with masked language modeling and attribute classification jointly. The first objective forces RoBERTa to infill a span consistent with the semantics and attributes represented by hidden states, while the latter one steers the hidden states towards a desired attribute.

We experiment with two fundamental revision tasks, text simplification and formalization. In text simplification, our method surpassed the supervised baseline by 4.2 SARI score and unsupervised baseline 5.3 SARI score on Newsela-turk (Maddela et al., 2020). In text formalization, our approach outperforms all of the unsupervised baseline models in terms of content preservation and formality on GYAFC-fr (Rao and Tetreault, 2018). Ablation study is conducted to validate the design of each component in the model, through which we have following key findings: (1) representation optimization is essential to formality metrics; (2) infilling conditioned on hidden states helps preserve content; (3) our gradient-guided span selection contributes to both of them.¹

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